MAgent Theory 2.0

Communication Links:
Alignment, Codes, and Dissonance

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Status and summary

This is a working document for the first item of the future-work list. It is not part of the main manuscript and is not cited by it. It uses the main manuscript's conventions: one structure group G acts on the belief and model sectors through two representations, an agent's frame is $U_i \in G$, and statements are made at one context in reference coordinates unless they say otherwise. To keep the formulas short, the belief sector is written with $V = \mathbb{R}^K$ and $G \subseteq \mathrm{GL}(K, \mathbb{R})$ acting by matrix multiplication; the model sector is analogous with its own representation.

The motivating picture is the following. When agent j communicates a belief to agent i, most of what the belief means to j is lost: j projects it into a low-dimensional public package, a sentence, a drawing or a signal, and i rebuilds it inside its own, larger internal space using its own conceptual structure. In the manuscript's index convention, with receiver i and sender j, the chain is

$$q_j \longrightarrow (T_j)_{\#} q_j \longrightarrow \text{noise} \longrightarrow \text{decoding by } i \longrightarrow q_i'. \tag{1}$$

Two different objects are built from the same maps. The composite of a sending and a receiving map has rank at most the width of the message space and is lossy. An exponentiated link with a generator of the same form is invertible and loses nothing. This document keeps them apart: each directed edge carries an invertible dictionary, which is the gauge field, and each modality carries a lossy code.

The main results are these. The receiver's decoder is not a free map: for linear Gaussian codes it is the gain computed from the receiver's own prior covariance, and the reconstructed belief adopts the sender's belief along the communicated directions while keeping the receiver's own belief along the discarded ones (proposition 2.3). For general laws and a noiseless code, the meaning lost in transmission equals the total disagreement minus the disagreement visible through the code (proposition 2.2). Coded messages identify the dictionary only up to the subgroup that leaves the code unchanged, a subgroup of dimension $K(K-r)$ in $\mathrm{GL}(K, \mathbb{R})$ for a code of rank r (theorem 3.1); several modalities sharing one dictionary identify it up to the intersection of their stabilizers, so overlapping modalities help less than independent ones (corollary 4.1). Exact consensus of transported beliefs is possible exactly on the common fixed space of the holonomies of the communication graph, so nontrivial holonomy on a loop is an obstruction to agreement, a dissonance (theorem 5.1). Finally, with n coded messages per exchange the posterior over who was heard behaves as a Gibbs row at temperature $1/n$ whose costs are divergences between message laws, not between full beliefs (proposition 6.1). Learning the dictionaries from grounded messages under a prior that penalizes holonomy concentrates the recognition law on the identified coset, and within it the loop term fills exactly those invisible directions that create holonomy (theorem 7.2). These are the same directions in which a true dissonance would go undetected and be absorbed into the dictionaries (proposition 7.4). A pair detects its mutual misunderstanding only in what both can say and repairs its dictionaries through reciprocity in what at least one can say (proposition 7.7), and loops through third agents can fix directions that no pair can (proposition 7.5 and corollary 7.6).

The theory is meant for general probability laws, and Gaussian laws are a computational simplification in this document. The results split as follows. The meaning-loss identity, the noisy-code bound, the bandwidth proposition, the query-map proposition and the consensus theorem hold for general laws; identifiability, the modality and routing corollaries, the structural parts of the learning chapter and the exact identification results hold for every location family of noise (remark 3.2); the decoder as a Kalman gain, the Gaussian reconstruction, and the limit laws, shrinkage formulas and receiver-owned variances of the learning chapter are Gaussian, and their linearity, data-independent covariances and exact factorized means are features of that family. Extensions to general laws are posed in open problems 8.1 and 8.4.

Every identity was also checked numerically at small sizes; the checks support the instances tested and are not proofs.

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Chapter 1

Codes and dictionaries

Definition 1.1 (Code of a modality). A modality c, such as speech, drawing or gesture, has a public message space $H_c = \mathbb{R}^{r_c}$ and a noise covariance R_c, either positive definite or zero for a noiseless code. The message space carries no frame of its own: its coordinates are public conventions. Sender j's code in modality c is a linear map $\bar{T}^c_j : V \to H_c$ of rank $r_c \leq K$ on reference coordinates. Its displayed form is $T^c_j = \bar{T}^c_j U_j^{-1}$. A message produced from displayed content y_j is

$$h = T^c_j y_j + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, R_c). \tag{1.1}$$

Under a passive change $U_j \mapsto g_j U_j$ the displayed content becomes $g_j y_j$ and the displayed code becomes $T^c_j g_j^{-1}$, so the message is unchanged. Public messages therefore do not depend on the sender's frame, and the reference code $\bar{T}^c_j$ is a frame-independent property of the sender. Codes are agent specific. A shared language is part of a shared generative model in the sense of the main manuscript: members of a species use codes related by the frame map of their shared model.

Definition 1.2 (Dictionary). Receiver i's dictionary for sender j is

$$\theta_{ij} = U_i \exp(\omega_{ij}) U_j^{-1} \in G, \qquad \bar{\theta}_{ij} = \exp(\omega_{ij}), \qquad \omega_{ij} \in \mathfrak{g}. \tag{1.2}$$

It is owned by the receiver: it is i's hypothesis about how j's internal reference coordinates sit relative to its own. Content with reference coordinates $\bar{y}$ for i corresponds, according to i, to $\bar{\theta}_{ij}^{-1}\bar{y}$ for j. The effective code that i attributes to j is therefore

$$\hat{A}^c_{ij} = \bar{T}^c_j \bar{\theta}_{ij}^{-1}. \tag{1.3}$$

The dictionary θ_{ji} is j's hypothesis about i. It is an independent datum, learned by a different agent, so θ_{ji} need not equal θ_{ij}^{-1}. The product $\bar{\theta}_{ij}\bar{\theta}_{ji}$ is the holonomy of the two-cycle $i \to j \to i$: the composite of what i thinks j means with what j thinks i means. It is the identity exactly when the two agents' translations of each other are mutually inverse; otherwise it measures mutual misunderstanding. The main manuscript's decision not to impose the inverse convention is, in this reading, the statement that two agents learn their dictionaries of each other separately.

Remark 1.3 (Why the dictionary is group valued). A model that i holds of j need not in general be a group element, and the group-valued form is a hypothesis with a stated scope. Four considerations support it within a species. First, the two agents' internal spaces are copies of one G-space, and the relative position of two frames of one principal bundle at a context is a group element; partial access is carried by the code, not by the dictionary. Second, one dictionary serves every modality, which is what lets several codes pool their evidence in corollary 4.1. Third, invertibility is used: the consensus theorem theorem 5.1 needs inverses to show that transported

beliefs are well defined along different paths, and holonomy has conjugacy-class invariants only for invertible links. Fourth, i's knowledge of j is a probability law over G, a recognition law of ω_{ij}, not a point; uncertainty about the dictionary is carried by that law. For agents of different kinds, whose internal spaces differ in dimension or representation, an invertible dictionary does not exist. The natural replacement is a linear or Markov map between different spaces, with consistency measured by global sections of a sheaf rather than by holonomy [9].

Chapter 2

Reception through a code

Fix a receiver i, a sender j and one modality, and suppress the indices. In i's reference coordinates let p be i's prior law of the content and let $q = (\bar{\theta})_{\#}\bar{q}_j$ be the sender's belief carried into i's coordinates by the dictionary. Write $A = \hat{A}$ for the effective code.

Proposition 2.1 (Reception of one message). Let $p = \mathcal{N}(m, \Sigma_p)$ with Σ_p positive definite, let A have full row rank r, and let R be positive definite. Given a message h from the model $h = A\bar{y} + \varepsilon$, the posterior of $\bar{y}$ is $\mathcal{N}(m + O(h - Am), \Sigma_p - OA\Sigma_p)$ with

$$O = \Sigma_p A^T (A\Sigma_p A^T + R)^{-1}. \tag{2.1}$$

The posterior precision is $\Sigma_p^{-1} + A^T R^{-1} A$, a correction of rank at most r, and the information carried by one message is

$$I(\bar{y}; h) = \tfrac{1}{2} \log\det\big(I_r + R^{-1} A\Sigma_p A^T\big). \tag{2.2}$$

Proof. The pair $(\bar{y}, h)$ is jointly Gaussian with $\mathrm{Cov}(\bar{y}, h) = \Sigma_p A^T$ and $\mathrm{Cov}(h) = A\Sigma_p A^T + R$, and Gaussian conditioning gives the posterior [12]. The precision form follows from the Woodbury identity. The mutual information is the difference of the entropies of $\bar{y}$ and $\bar{y}$ given h, $\frac{1}{2}\log\det\Sigma_p - \frac{1}{2}\log\det(\Sigma_p - OA\Sigma_p)$, and the matrix determinant lemma turns this into eq. (2.2). □

The receiving map is therefore derived. It is the gain O, fixed by the receiver's own uncertainty Σ_p and by its model A of the sender's code. The rank bound makes precise the loss in the motivating picture: one message changes the receiver's precision in at most r directions, however large K is.

The proposition is the Gaussian instance of a general fact: for any prior law and any message kernel, the receiver's decoder is the conditional law of the content given the message, which is the reconstruction eq. (2.3) below in the noiseless case. That the decoder is a linear map, that the posterior covariance does not depend on the message received, and that one message changes the precision in at most r directions are features of the Gaussian family, not of decoding in general.

The next statement concerns the reconstruction of a whole belief rather than the response to one message. For a measurable code T and laws q, p on a standard Borel space, choose a regular conditional law $p(dy \mid h)$ of p given $T = h$, and define the reconstruction

$$\hat{q}(dy) = \int p(dy \mid h)\,(T_{\#}q)(dh). \tag{2.3}$$

It keeps exactly what the code transmits, the message law $T_{\#}q$, and fills every discarded direction with the receiver's own conditional belief.

Proposition 2.2 (Meaning lost through a noiseless code). If $D_{\mathrm{KL}}(q\|p) < \infty$, then

$$D_{\mathrm{KL}}(q\|\hat{q}) = D_{\mathrm{KL}}(q\|p) - D_{\mathrm{KL}}(T_{\#}q\|T_{\#}p). \tag{2.4}$$

Proof. For laws on standard Borel spaces and a measurable map T, the chain rule for relative entropy gives

$$D_{\mathrm{KL}}(\mu\|\nu) = D_{\mathrm{KL}}(T_{\#}\mu\|T_{\#}\nu) + \int D_{\mathrm{KL}}\big(\mu(\cdot \mid h)\|\nu(\cdot \mid h)\big)(T_{\#}\mu)(dh) \tag{2.5}$$

in $[0,\infty]$, with regular conditional laws given T [6, 11]. Apply it to (q,p). Apply it also to $(q,\hat{q})$: by construction $T_{\#}\hat{q} = T_{\#}q$ and $\hat{q}(\cdot \mid h) = p(\cdot \mid h)$ for $T_{\#}q$-almost every h, so the first term vanishes and the second is the same integral. Since $D_{\mathrm{KL}}(T_{\#}q\|T_{\#}p) \le D_{\mathrm{KL}}(q\|p) < \infty$, subtraction is legitimate. □

In words, the meaning lost when a belief is sent through a code and rebuilt by a receiver equals the full disagreement between sender and receiver minus the disagreement that the code makes visible. It vanishes exactly when the code is sufficient for the pair, that is, when the sender's conditional belief given the message already equals the receiver's.

Proposition 2.3 (Gaussian reconstruction). Let $q = \mathcal{N}(a,\Sigma_q)$, $p = \mathcal{N}(m,\Sigma_p)$ with positive-definite covariances, and let A have full row rank. Put

$$G = \Sigma_p A^T (A\Sigma_p A^T)^{-1}, \qquad P = GA. \tag{2.6}$$

Then $AG = I_r$, $P^2 = P$, $\ker P = \ker A$, the range of P is the range of $\Sigma_p A^T$, and the reconstruction through the noiseless code A is

$$\hat{q} = \mathcal{N}\big((I-P)m + Pa,\ (I-P)\Sigma_p(I-P)^T + P\Sigma_q P^T\big). \tag{2.7}$$

Proof. The identities for G and P are immediate. Conditioning p on $A\bar{y} = h$ gives $\mathcal{N}(m + G(h - Am), \Sigma_p - GA\Sigma_p)$. Mixing over $h \sim \mathcal{N}(Aa, A\Sigma_q A^T)$ gives mean $m + GA(a-m) = (I-P)m + Pa$ and covariance $\Sigma_p - P\Sigma_p + GA\Sigma_q A^T G^T$. The last term is $P\Sigma_q P^T$. The matrix $P\Sigma_p = \Sigma_p A^T (A\Sigma_p A^T)^{-1} A\Sigma_p$ is symmetric and satisfies $P\Sigma_p P^T = P\Sigma_p$, so $\Sigma_p - P\Sigma_p = (I-P)\Sigma_p(I-P)^T$. □

The reconstructed belief adopts the sender's belief along the communicated directions and keeps the receiver's own belief along the discarded ones. Its update of the mean,

$$\mu_i' = \mu_i + G\hat{A}_{ij}\big(\bar{\theta}_{ij}\bar{\mu}_j - \mu_i\big), \tag{2.8}$$

is a projected message of the kind computed by attention. The direction in which a communicated coordinate is lifted, the range of $\Sigma_p A^T$, is set by the receiver's own correlations: two receivers with different priors read the same message as meaning different things. This is the formal content of the statement that the receiver internalizes the message through its own conceptual structure. The loss eq. (2.4) evaluates in closed form for eq. (2.7).

Corollary 2.4 (Noisy codes). Let a noisy code be a Markov kernel $\mathcal{K}$ from the content space to a standard Borel message space, for example the Gaussian kernel $\mathcal{K}(\bar{y}, \cdot) = \mathcal{N}(A\bar{y}, R)$. Let p^h and q^h be regular versions of the posteriors of p and q given h, and define the reconstruction $\hat{q} = \int p^h \, (q\mathcal{K})(dh)$. If $D_{\mathrm{KL}}(q\|p) < \infty$, then

$$D_{\mathrm{KL}}(q\|\hat{q}) \leq D_{\mathrm{KL}}(q\|p) - D_{\mathrm{KL}}(q\mathcal{K}\|p\mathcal{K}). \tag{2.9}$$

Proof. The chain rule applied to the joint laws of $(\bar{y}, h)$ under $q \otimes \mathcal{K}$ and $p \otimes \mathcal{K}$, which share the kernel, gives $D_{\mathrm{KL}}(q\|p) = D_{\mathrm{KL}}(q\mathcal{K}\|p\mathcal{K}) + \int D_{\mathrm{KL}}(q^h\|p^h)(q\mathcal{K})(dh)$. Since $q = \int q^h(q\mathcal{K})(dh)$, joint convexity of relative entropy bounds $D_{\mathrm{KL}}(q\|\hat{q})$ by the integral. □

The proof uses only the chain rule and joint convexity, so the bound holds for every kernel and every pair of laws with $D_{\mathrm{KL}}(q\|p) < \infty$. For Gaussian laws and the Gaussian kernel, $\hat{q}$ has mean $m + OA(a - m)$ and covariance $\Sigma_p - OA\Sigma_p + O(A\Sigma_q A^T + R)O^T$ with O from eq. (2.1); at the tested instance the bound holds with slack 3.4×10^{-3}.

Remark 2.5 (The link with a code-shaped generator). For matrices O of size $K \times r$ and T of size $r \times K$, $e^{OT} = I + O\varphi_1(TO)T$ with $\varphi_1(z) = (e^z - 1)/z$ [10, Section 1.8], and $\det e^{OT} = e^{\mathrm{tr}(OT)} > 0$. With the gain of eq. (2.6), $AG = I_r$ gives $e^P = I + (e - 1)P$: the invertible link whose generator is the reconstruction projector is a dilation along the communicated directions, not the reconstruction. Reconstruction is the affine use $(I - P)m + Pa$ of the projector, and it is not invertible. This is the precise sense in which no group element factors through a lossy code and the two are related only through the Lie algebra.

Chapter 3

What messages reveal about the dictionary

A dictionary can be tested only against something both agents refer to. Let a referent have law ρ on V in i's reference coordinates, with positive-definite second moment $M = \mathbb{E}_\rho[\bar{y}\bar{y}^T]$. The sender encodes the referent in its own coordinates, and the receiver models the message as

$$h \mid \bar{y} \sim \mathcal{N}\big(\bar{T}\bar{\theta}^{-1}\bar{y}, R\big), \tag{3.1}$$

with R positive definite.

Theorem 3.1 (Identifiability up to the code stabilizer). Let $\mathrm{Stab}_G(\bar{T}) = \{g \in G : \bar{T}g = \bar{T}\}$.

(i) Two dictionaries $\bar{\theta}', \bar{\theta}''$ give the same conditional message law for every referent if and only if $\bar{\theta}'^{-1}\bar{\theta}'' \in \mathrm{Stab}_G(\bar{T})$.

(ii) For the perturbation $\bar{\theta}e^{\delta}$, $\delta \in \mathfrak{g}$, the Fisher information of eq. (3.1), averaged over ρ, is

$$F(\delta, \delta) = \mathrm{tr}\big(R^{-1}\bar{T}\delta C \delta^T \bar{T}^T\big), \qquad C = \bar{\theta}^{-1} M \bar{\theta}^{-T}, \tag{3.2}$$

and its kernel is $\{\delta \in \mathfrak{g} : \bar{T}\delta = 0\}$, the Lie algebra of $\mathrm{Stab}_G(\bar{T})$.

(iii) For $G = \mathrm{GL}(K, \mathbb{R})$ and $\bar{T}$ of rank r, $\mathrm{Stab}_G(\bar{T})$ has dimension $K(K-r)$, and the identifiable part of the dictionary has dimension Kr.

Proof. (i) Gaussian laws with a common covariance agree exactly when their means agree, and $\bar{T}\bar{\theta}'^{-1}\bar{y} = \bar{T}\bar{\theta}''^{-1}\bar{y}$ for every $\bar{y}$ means $\bar{T}\bar{\theta}'^{-1} = \bar{T}\bar{\theta}''^{-1}$, that is, $\bar{T}\bar{\theta}'^{-1}\bar{\theta}'' = \bar{T}$. The set $\mathrm{Stab}_G(\bar{T})$ is a subgroup because $\bar{T}g_1g_2 = \bar{T}g_2 = \bar{T}$. (ii) The mean $\bar{T}e^{-\delta}\bar{\theta}^{-1}\bar{y}$ has derivative $-\bar{T}\delta\bar{\theta}^{-1}\bar{y}$ at $\delta = 0$, and the Fisher information of a Gaussian mean with fixed covariance R is $\mathbb{E}_\rho[(\bar{T}\delta\bar{\theta}^{-1}\bar{y})^T R^{-1}(\bar{T}\delta\bar{\theta}^{-1}\bar{y})]$, which is eq. (3.2). Since C and R^{-1} are positive definite, $F(\delta,\delta) = 0$ exactly when $R^{-1/2}\bar{T}\delta C^{1/2} = 0$, that is, when $\bar{T}\delta = 0$. For such δ every power of δ is annihilated by $\bar{T}$, so $\bar{T}e^{t\delta} = \bar{T}$ for all t. (iii) $\bar{T}g = \bar{T}$ means that $g - I$ maps V into $\ker\bar{T}$, which has dimension $K - r$; such g form an open subset of an affine space of dimension $K(K-r)$ containing the identity. □

Messages never test the part of the dictionary that the code discards. Relative to the observable algebra of coded messages that part is gauge in the sense of the main manuscript's definition of active transformations: no message distinguishes two dictionaries that differ by an element of the stabilizer. This is the precise form of the statement that words cannot carry what a belief means to its holder. Without shared referents nothing is identified: if the receiver had only the mean of ρ, the data would determine $\bar{T}\bar{\theta}^{-1}$ only on one vector. Noise rescales the Fisher information but does not change its kernel. At $K = 5$ and $r = 2$ the numerical check gives Fisher rank $10 = Kr$ out of 25.

Remark 3.2 (Noise laws beyond the Gaussian). Gaussian noise is a computational simplification here. Let the grounded model be the location family $h = \bar{T}\bar{\theta}^{-1}\bar{y} + \varepsilon$ with ε independent of $\bar{y}$ and of a fixed law ν on $\mathbb{R}^r$. Parts (i) and (iii) of theorem 3.1 hold for every ν, because a probability law on $\mathbb{R}^r$ differs from each of its nonzero translates: its characteristic function φ is nonzero near the origin, and $e^{\mathrm{i}\langle t,a\rangle}\varphi(t) = \varphi(t)$ there forces $a = 0$. If ν has a continuously differentiable density f with finite Fisher information $J_f = \mathbb{E}[\nabla \log f(\varepsilon)\nabla \log f(\varepsilon)^T]$, positive definite, then part (ii) holds with R^{-1} replaced by J_f,

$$F(\delta, \delta) = \operatorname{tr}\big(J_f \bar{T}\delta C \delta^T \bar{T}^T\big), \tag{3.3}$$

with the same kernel, since the score of the perturbation is $\nabla \log f(\varepsilon)^T \bar{T}\delta\bar{\theta}^{-1}\bar{y}$ and ε is independent of $\bar{y}$. Corollaries 4.1 and 4.2 and the structural results of chapter 7 therefore hold for every such noise law. The Gaussian value $J_f = R^{-1}$ is the instance with the least Fisher information at a given covariance, by the Cramér–Rao inequality for location families. In the numerical check with independent Student-t noise of five degrees of freedom, a Monte Carlo estimate of the Fisher form agrees with eq. (3.3) to 0.3 percent. Codes whose messages are not a location family, such as codes into finite alphabets, are open problem 8.4.

Chapter 4

Several modalities

Corollary 4.1 (Pooling modalities through one dictionary). Let modalities $c = 1, \dots, C$ have codes $\bar{T}^c$ and noises R_c, all read through the same dictionary $\bar{\theta}$. Then the dictionary is identified up to $\bigcap_c \mathrm{Stab}_G(\bar{T}^c)$, and the Fisher information is $\sum_c F_c$ with kernel $\{\delta \in \mathfrak{g} : \bar{T}^c\delta = 0 \text{ for all } c\}$. For $G = \mathrm{GL}(K, \mathbb{R})$ the identifiable dimension is

$$K \dim\Big(\sum_c \operatorname{row} \bar{T}^c\Big) \le K \min\Big(K, \sum_c r_c\Big), \tag{4.1}$$

with equality on the right exactly when the row spaces of the codes are independent, and the dictionary is fully identified exactly when the row spaces span the dual of V.

Proof. Identity of the conditional laws of all messages requires identity for each modality, which gives the intersection by theorem 3.1. Fisher information adds over conditionally independent messages, and a sum of positive-semidefinite forms vanishes exactly where each does. For $\mathrm{GL}(K, \mathbb{R})$, $\bar{T}^c\delta = 0$ for all c means that the range of δ lies in $\bigcap_c \ker \bar{T}^c$, whose dimension is K minus the dimension of the sum of the row spaces. □

Modalities that overlap, in the sense that their row spaces share directions, add less than their widths suggest; complementary modalities, such as speech and drawing when they convey different aspects, add fully. The numerical check uses two codes of widths 2 and 2 that share one direction: the identifiable dimension is $15 = 5 \times 3$, not 20. The corollary depends on the dictionary being shared across modalities. Without it each modality would have its own effective code to learn, and the modalities could not pool their evidence about the other agent.

Corollary 4.1 assumes that receiver i hears sender j in every modality. A receiver with several senders instead chooses, separately in each modality, whom to listen to. Let $\beta_{ij}^{(c)}$ be the probability that modality c of receiver i carries a message from j, and let $F_c^{(j)}$ denote the form eq. (3.2) for the code $\bar{T}_j^c$, the noise R_c and the dictionary $\bar{\theta}_{ij}$.

Corollary 4.2 (Routed modalities). In each exchange let a referent $\bar{y} \sim \rho$ be drawn, and for each modality c independently let a source label J_c be drawn with $\mathbb{P}(J_c = k) = \beta_{ik}^{(c)}$ and a message with law $\mathcal{N}(\bar{T}_k^c \bar{\theta}_{ik}^{-1} \bar{y}, R_c)$ given $J_c = k$ and $\bar{y}$. Consider the dictionary $\bar{\theta}_{ij} e^{\delta}$ of one edge $j \to i$.

(i) If the labels are observed, the Fisher information about δ per exchange is $\sum_c \beta_{ij}^{(c)} F_c^{(j)}$.

(ii) If the labels are latent, the Fisher information is at most $\sum_c \beta_{ij}^{(c)} F_c^{(j)}$ in the Loewner order.

(iii) With observed labels the information vanishes exactly on $\{\delta \in \mathfrak{g} : \bar{T}_j^c \delta = 0 \text{ for every } c \text{ with } \beta_{ij}^{(c)} > 0\}$.

Proof. (i) Given $\bar{y}$ and the labels, the messages of different modalities are independent, so the complete-data log likelihood is a sum over c. A term with $J_c = k \neq j$ does not depend on $\bar{\theta}_{ij}$ and contributes nothing. A term with $J_c = j$ contributes the information of theorem 3.1. Averaging the indicator of $J_c = j$ gives the weight $\beta_{ij}^{(c)}$. (ii) By the missing-information principle [14], the information of the observed data equals the complete-data information minus the expected information about δ carried by the conditional law of the labels given the messages, which is positive semidefinite. (iii) A sum of positive-semidefinite forms with positive weights vanishes exactly where every form vanishes, and theorem 3.1 identifies the kernel of each. □

A modality that never listens to j teaches i nothing about its dictionary for j, and the directions that the attending modalities' codes discard stay invisible. When the receiver must infer who was heard, it learns less than the routing-weighted sum. In the numerical check, with two modalities attending to the target sender with probabilities 0.7 and 0.2, a Monte Carlo estimate of the complete-data information agrees with $\sum_c \beta_{ij}^{(c)} F_c^{(j)}$ to 0.6 percent, and the latent-label information is smaller in the Loewner order.

Proposition 4.3 (A receiver's reading of a code is a dictionary modulo the stabilizer). Let $\bar{T}$ and S be $r \times K$ matrices of full row rank r. Then $\{\theta \in \mathrm{GL}(K, \mathbb{R}) : \bar{T}\theta^{-1} = S\}$ is a single left coset $\theta_0 \, \mathrm{Stab}_{\mathrm{GL}(K,\mathbb{R})}(\bar{T})$, and for $r < K$ it meets $\mathrm{GL}^+(K, \mathbb{R})$.

Proof. Complete the rows of $\bar{T}$ to a basis of the dual space and let P be the invertible matrix whose first r rows are those of $\bar{T}$; define Q likewise from S. With $E = (I_r \; 0)$, $\bar{T} = EP$ and $S = EQ$, so $\theta_0^{-1} = P^{-1}Q$ satisfies $\bar{T}\theta_0^{-1} = S$. For $r < K$, changing the sign of a completing row of Q changes the sign of $\det \theta_0$ without changing EQ. Finally $\bar{T}\theta^{-1} = \bar{T}\theta_0^{-1}$ exactly when $\bar{T}\theta_0^{-1}\theta = \bar{T}$, that is, when $\theta_0^{-1}\theta \in \mathrm{Stab}(\bar{T})$. □

A receiver that reads a sender's code through an arbitrary full-rank map S therefore holds exactly the identifiable part of a dictionary, a point of the rK-dimensional quotient of theorem 3.1. The proposition also applies to the stacked code $(\bar{T}^1; \ldots; \bar{T}^C)$ and the stacked readings $(S_1; \ldots; S_C)$. When both stacks have full row rank $\sum_c r_c \leq K$, there is one dictionary $\bar{\theta}$ with $S_c = \bar{T}^c\bar{\theta}^{-1}$ for every c, unique modulo $\bigcap_c \mathrm{Stab}(\bar{T}^c)$: readings learned independently for each modality are then automatically readings of one shared dictionary. Sharing constrains the readings only when the total width exceeds K or the two stacks have different ranks. It constrains them substantively when one dictionary must serve two roles, as in remark 4.4.

Remark 4.4 (Attention heads as modalities). Multi-head attention [19] has the structure of routed modalities. The correspondence is collected below, with modality c written as head h.

Communication links	Multi-head attention
modality c	head h
sender's code $\bar{T}_j^c$ into H_c of width r_c	key map, and the value map that sends content, of width d_h
receiver's reading $S_c = \bar{T}^c\bar{\theta}^{-1}$ of the code	query map
decoder, the gain of proposition 2.1	output map of the head
routing $\beta_{ij}^{(c)}$	attention weights of head h
frames U_i	per-token frames, when present

Standard attention is a special case of the construction here in three respects. All tokens share every code, which is one species with one language. The output map is learned rather than derived from the receiver's covariance. The query and key maps are shared by all tokens, so the dictionary is the same on every edge; a closed walk of length k then has holonomy $\bar{\theta}^k$ in reference coordinates, the self-loop included, so holonomy can be nontrivial but cannot depend on which edges a loop uses. When the heads' total width is K and the stacked maps are invertible, the stacked form of proposition 4.3 shows that the independently learned query maps are readings of one shared dictionary, so the heads already pool their evidence as in corollaries 4.1 and 4.2; the stacked code is then invertible, the condition for full identification. What standard attention does not impose is one dictionary for routing and content. The query and key maps decide whom to listen to, the value and output maps carry and decode the content, and the two are learned independently. Requiring the receiver to use the same dictionary both to choose its sources and to interpret what it hears, with the decoder derived from that reading as in proposition 2.1, is a substantive constraint. A readout in which the key map is also the sending map and the output map is the gain built from the query map satisfies it.

Chapter 5

Curvature as dissonance

Let the communication graph Γ have a directed edge $j \to i$ when i receives from j, with dictionaries $\bar{\theta}_{ij}$. For a directed closed walk $\gamma = (i_0 \to i_1 \to \cdots \to i_k = i_0)$ define its holonomy $H_\gamma = \bar{\theta}_{i_k i_{k-1}} \cdots \bar{\theta}_{i_1 i_0}$, the composite of the receivers' dictionaries along the walk. Reference belief means $(\mu_i)_i$ are in consensus when

$$\mu_i = \bar{\theta}_{ij}\mu_j \qquad \text{for every edge } j \to i, \tag{5.1}$$

so that every receiver, translating each sender's belief through its dictionary, finds agreement.

Theorem 5.1 (Consensus is the fixed space of holonomy). Let Γ be strongly connected and fix i_0. The map $(\mu_i)_i \mapsto \mu_{i_0}$ is a linear bijection from the solutions of eq. (5.1) onto $\bigcap_\gamma \mathrm{Fix}(H_\gamma)$, the intersection over directed closed walks based at i_0 of the fixed spaces of their holonomies.

Proof. If (μ_i) solves eq. (5.1), composing the edge relations along γ gives $\mu_{i_0} = H_\gamma \mu_{i_0}$. Conversely let v be fixed by every H_γ. For each i choose a directed walk p from i_0 to i and set $\mu_i = \Pi(p)v$, where $\Pi(p)$ is the composite of dictionaries along p. If p' is another such walk, choose a directed walk s from i back to i_0, which exists by strong connectivity. Then $\Pi(s)\Pi(p)$ and $\Pi(s)\Pi(p')$ are holonomies of closed walks at i_0, so both fix v, and since $\Pi(s)$ is invertible, $\Pi(p)v = \Pi(p')v$. The definition is therefore independent of the walk, and appending an edge $j \to i$ to a walk to j gives $\mu_i = \bar{\theta}_{ij}\mu_j$. The two constructions are mutually inverse and linear. □

The proof uses the invertibility of the dictionaries, which is one of the reasons recorded in remark 1.3. A directed closed walk of length two is admitted, so the two-cycle holonomy $\bar{\theta}_{ij}\bar{\theta}_{ji}$ constrains consensus as much as any longer loop. If the holonomies have no common nonzero fixed vector, the only consensus is the zero mean in reference coordinates, which is a convention rather than an agreement; the community cannot share a common reference, although each pair can communicate.

The theorem is stated for means, but the same proof applies to laws of any kind. Transported laws are in consensus, $q_i = (\bar{\theta}_{ij})_\# q_j$ on every edge, exactly when q_{i_0} is invariant under the pushforward by the holonomy of every closed walk at i_0: compose pushforwards along walks instead of matrices. For Gaussian laws this constrains both the mean and the covariance; for general laws it constrains every feature of q_{i_0} that the holonomies move.

Corollary 5.2 (Frustration). For positive weights w_{ij}, the consensus energy $E(\mu) = \sum_{j \to i} w_{ij} \|\mu_i - \bar{\theta}_{ij}\mu_j\|^2$ is a positive-semidefinite quadratic form whose kernel is the solution set of eq. (5.1). Its least eigenvalue on configurations of unit norm is positive exactly when the holonomies have no common nonzero fixed vector.

Proof. $E(\mu) = \|B\mu\|^2$ for the weighted incidence operator B with blocks $\sqrt{w_{ij}}(I, -\bar{\theta}_{ij})$, so $E = \mu^T B^T B \mu$ and its kernel is $\ker B$. The least eigenvalue of $B^T B$ is positive exactly when $\ker B = 0$, and theorem 5.1 identifies $\ker B$. □

For orthogonal links on undirected graphs with the inverse convention this is the connection Laplacian, and its least eigenvalue as a measure of frustration was studied by Singer and Wu [16] and Bandeira et al. [2]. The directed version above admits independent opposite dictionaries. In the numerical check, three agents with planar rotation dictionaries whose triangle holonomy is the identity have a two-dimensional consensus space; with holonomy angle 0.7 the consensus space is zero and the least eigenvalue is 0.054. Two agents with mutually inverse dictionaries have a two-dimensional consensus space; with two-cycle holonomy angle 0.3 it is zero.

Definition 5.3 (Dissonance of a loop). The dissonance of a directed closed walk γ is the conjugacy class of H_γ. A plaquette, a shortest cycle of Γ, is consonant when its holonomy is the identity and dissonant otherwise. A quantitative label is any conjugation-invariant defect of H_γ, for example $\|H_\gamma - I\|_F$ for orthogonal dictionaries; for general dictionaries the main manuscript's law-dressed loop defect supplies an invariant measure.

Two limits on what dissonance can affect follow from the earlier chapters. Agents test agreement only through codes, so the consensus they can detect is the weaker condition $\bar{T}_j^c(\mu_j - \bar{\theta}_{ij}^{-1}\mu_i) = 0$ for every edge and modality, and dissonance in directions that no code reads is never detected; proposition 7.4 makes this precise for learned dictionaries. And a law that is invariant under part of a holonomy, such as an isotropic belief under a rotation, cannot register that part; the main manuscript's loop-blindness proposition states this at the level of laws.

Chapter 6

Bandwidth and source rows

Proposition 6.1 (Coded messages give Gibbs rows at temperature one over the message count). Let receiver i consider finitely many candidate sources j with prior $\pi_j > 0$, each modeled as emitting independent messages with density ν_j on the message space. Let $h_1, h_2, \dots$ be independent with density ν, and suppose $\mathbb{E}_\nu |\log \nu_j| < \infty$ for every j. Then, almost surely,

$$\frac{1}{n} \log \frac{\mathbb{P}(J = j \mid h_{1:n})}{\mathbb{P}(J = k \mid h_{1:n})} \longrightarrow D_{\mathrm{KL}}(\nu \| \nu_k) - D_{\mathrm{KL}}(\nu \| \nu_j). \tag{6.1}$$

Proof. By Bayes' rule the left side is $\frac{1}{n} \log(\pi_j / \pi_k) + \frac{1}{n} \sum_{t \le n} [\log \nu_j(h_t) - \log \nu_k(h_t)]$. The strong law of large numbers sends the average to $\mathbb{E}_\nu \log \nu_j - \mathbb{E}_\nu \log \nu_k$, which equals the right side because the entropy of ν cancels. □

To leading exponential order, the posterior over who was heard is the Gibbs row $\pi_j e^{-D_{\mathrm{KL}}(\nu \| \nu_j)/T} / Z$ at temperature $T = 1/n$, the posterior concentration of a possibly misspecified model [4]. The costs are divergences between message laws, which are code-visible divergences: by data processing they are at most the corresponding divergences between full beliefs, with equality exactly when the code is sufficient. The law-level peer costs of the main manuscript's reciprocal potential therefore arise in the limit of many messages through an injective, noiseless code; its registered-agreement theorem gives an exact finite-n version for type registration. Finite bandwidth means both a finite temperature and a selective cost.

Chapter 7

Learning the dictionaries

The receiver owns its dictionaries and has to learn them. Theorem 3.1 says what messages can teach: a coset of the stabilizer of the code. This chapter asks what a prior that rewards agreement around loops adds to that, and answers the question exactly in the tangent model, the linearization at the identity dictionaries. The identity is the natural prior mean. Before any exchange the receiver assumes that the sender means what it would mean itself.

The tangent model is a computational simplification in two respects: it linearizes at the identity dictionaries, and it takes Gaussian noise and a Gaussian anchor. Its conclusions split accordingly. The invisible directions, the unreachable directions and their sheaf count, the split of the loop space into filled and detectable directions as a dimension count, and the exact identification results propositions 7.8 and 7.9 depend only on the identifiability kernel and on the loops; by remark 3.2 they hold for every location family of noise and every prior. The limiting law as a conditioned Gaussian, the shrinkage and bias formulas, and the variances of remark 7.10 are features of the Gaussian family.

7.1 The tangent model

Definition 7.1 (Tangent learning model). Let Γ be a finite communication graph with vertex set $\mathcal{V}$ and edge set $\mathcal{E}$, and let the edge $e : j \to i$ carry receiver i's generator $\omega_e = \omega_{ij} \in \mathfrak{g} \subseteq \mathfrak{gl}(K, \mathbb{R})$, so that $\omega = (\omega_e)_e \in X = \mathfrak{g}^{\mathcal{E}}$. Write $\bar{T}_e$ for the stacked code of the modalities with $\beta_{ij}^{(c)} > 0$ in corollary 4.2. Let $\mathcal{L}$ be a finite set of directed closed walks of Γ that contains every two-cycle, and define the linearized holonomy $B : X \to \mathfrak{g}^{\mathcal{L}}$ by $(B\omega)_\gamma = \sum_{e \in \gamma} \omega_e$, an edge being counted as often as γ traverses it. The tangent learning model consists of the prior law with density proportional to

$$\exp\Big(-\tfrac{1}{2}\omega^T \Pi_0 \omega - \tfrac{\lambda}{2} \sum_{\gamma \in \mathcal{L}} \|(B\omega)_\gamma\|^2_{W_\gamma}\Big), \tag{7.1}$$

where Π_0 is positive definite on X, $\lambda \geq 0$, and each W_γ is a positive-definite quadratic form on $\mathfrak{g}$, with $W = \bigoplus_\gamma W_\gamma$; and of the messages that i attributes to j, with observed labels as in corollary 4.2(i), each produced from a referent $\bar{y} \sim \rho$ with second moment M positive definite as

$$h = \bar{T}_e^c (I - \omega_e)\bar{y} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, R_c). \tag{7.2}$$

The prior is Gaussian with precision $\Lambda_0 = \Pi_0 + \lambda B^T W B$, which is positive definite, so the prior is normalized for every $\lambda \geq 0$. The message model is the first-order form of eq. (3.1) at $\bar{\theta}_e = e^{\omega_e}$, and $(B\omega)_\gamma$ is the first-order form of $\log H_\gamma$. By the Baker–Campbell–Hausdorff formula both errors are of second order in ω; for a fixed triangle the numerical check finds $|\log H_\gamma - (B\omega)_\gamma|/\epsilon^2 = 2.31$ at

$\epsilon = 10^{-2}$ and at $\epsilon = 10^{-3}$. Inside the tangent model the likelihood is Gaussian and linear in ω. The recognition law, here the exact posterior, is therefore Gaussian after any number of messages and is updated one message at a time by the information filter

$$\Lambda \leftarrow \Lambda + J_t^T R_c^{-1} J_t, \qquad \eta \leftarrow \eta + J_t^T R_c^{-1}\big(h_t - \bar{T}_e^c \bar{y}_t\big), \qquad J_t\omega = -\bar{T}_e^c \omega_e \bar{y}_t, \tag{7.3}$$

started at $(\Lambda_0, 0)$, with mean $\Lambda^{-1}\eta$ and covariance Λ^{-1}.

A change of the common reference coordinates by $h \in G$ acts by $\omega_e \mapsto h^{-1}\omega_e h$ under reference consistency, and then $(B\omega)_\gamma \mapsto h^{-1}(B\omega)_\gamma h$. The loop term is invariant when each W_γ is. The Frobenius norm is invariant under orthogonal h. For $\mathrm{GL}(K,\mathbb{R})$ the dressed norm $\|X\|_\Sigma^2 = \mathrm{tr}(X^T\Sigma^{-1}X\Sigma)$, with Σ the reference covariance of the belief of the agent at which γ is based, satisfies $\|h^{-1}Xh\|^2_{h^{-1}\Sigma h^{-T}} = \|X\|^2_\Sigma$. A law is needed here because $\mathfrak{gl}(K,\mathbb{R})$ carries no conjugation-invariant positive-definite form for $K \geq 2$: conjugation by $\mathrm{diag}(t,1,\dots,1)$ multiplies E_{12} by t. The anchor Π_0 need not be invariant, since it expresses the receiver's expectation in its own reference coordinates.

For the statements below, let $N_e = \{\delta \in \mathfrak{g} : \bar{T}_e\delta = 0\}$, the Lie algebra of the stabilizer of the stacked code, and let $N = \bigoplus_e N_e \subseteq X$, the invisible directions. Let Z be a matrix whose columns form an orthonormal basis of N for the Frobenius inner product on X, and Y one for $N^\perp$. Put

$$S = Z^T\Lambda_0 Z, \qquad \Pi_N = ZS^{-1}Z^T\Lambda_0, \qquad L_N = Z^T B^T W B Z, \tag{7.4}$$

so that Π_N is the Λ_0-orthogonal projection of X onto N and L_N is the loop operator on the invisible directions. Let $\bar{F}_e = \sum_c \beta_{ij}^{(c)} F_c^{(j)}$ be the information per exchange of corollary 4.2(i), evaluated at $\bar{\theta}_e = I$, where $C = M$.

7.2 Concentration and selection

Theorem 7.2 (Learning with a loop prior). In the tangent learning model let the truth be $\omega^\star \in X$, let each edge e receive $n_e = \lceil \kappa_e n \rceil$ independent exchanges with $\kappa_e > 0$, and write $z^\star = Z^T\omega^\star$.

(i) For every set of messages the likelihood is constant on each coset $\omega + N$, so the density of the recognition law relative to the prior depends on ω only through $Y^T\omega$.

(ii) Almost surely, as $n \to \infty$, the mean of the recognition law converges to

$$\omega_\infty = (I - \Pi_N)\,\omega^\star, \tag{7.5}$$

its covariance is $ZS^{-1}Z^T + O(1/n)$, and n times its covariance in the coordinates $Y^T\omega$ converges to $(Y^T F_\kappa Y)^{-1}$ with $F_\kappa = \bigoplus_e \kappa_e \bar{F}_e$. Equivalently, the recognition law converges to the prior conditioned on the coset $\omega^\star + N$. It concentrates on the coset at rate n in the identified directions, and within the coset it is the loop prior restricted to the coset.

(iii) Let $\Pi_0 = \sigma^{-2}I$. In the coordinates $Z^T\omega$ the limiting covariance is $\sigma^2(I + \lambda\sigma^2 L_N)^{-1}$. It equals the prior covariance on $\ker L_N = Z^T(N \cap \ker B)$, is strictly smaller on the orthogonal complement, and tends to zero there as $\lambda \to \infty$. The rank of L_N is $\dim B(N) = \dim N - \dim(N \cap \ker B)$.

(iv) If in addition the truth is flat, $B\omega^\star = 0$, then

$$Z^T(\omega_\infty - \omega^\star) = -(I + \lambda\sigma^2 L_N)^{-1} z^\star. \tag{7.6}$$

The loop term recovers the invisible part of the truth along an eigenvector of L_N with eigenvalue $\mu > 0$ up to the factor $\lambda\sigma^2\mu/(1+\lambda\sigma^2\mu)$, and as $\lambda \to \infty$ it recovers all of it except the component in $\ker L_N$, which stays at the prior mean zero.

(v) For a general truth, as $\lambda \to \infty$,

$$Z^T(\omega_\infty - \omega^\star) \longrightarrow -P_{\ker} z^\star - L_N^+ Z^T B^T W B \omega^\star, \tag{7.7}$$

with $P_{\ker}$ the orthogonal projection onto $\ker L_N$ and L_N^+ the pseudoinverse.

Proof. (i) By eq. (7.2) the law of a message on edge e depends on ω only through $\bar{T}_e^c \omega_e$ for attending modalities c, which does not change when ω_e changes by an element of N_e. Bayes' rule gives the second statement.

(ii) After the messages the precision is $\Lambda_n = \Lambda_0 + nF_n$ with $nF_n = \sum_t J_t^T R_c^{-1} J_t$, and since $h_t - \bar{T}_e^c \bar{y}_t = J_t\omega^\star + \varepsilon_t$, the information vector is $\eta_n = nF_n\omega^\star + \xi_n$ with $\xi_n = \sum_t J_t^T R_c^{-1}\varepsilon_t$. Every J_t vanishes on N, so $F_n = Y F_{Y,n} Y^T$ and $Z^T\xi_n = 0$. By the strong law of large numbers $F_{Y,n} \to Y^T F_\kappa Y$ almost surely, and this limit is positive definite because the kernel of F_κ is N by corollary 4.2(iii). In the basis (Y, Z) the matrix Λ_n has blocks $A_n = Y^T\Lambda_0 Y + nF_{Y,n}$, $D = Y^T\Lambda_0 Z$ and S, and block inversion gives

$$\Lambda_n^{-1} = ZS^{-1}Z^T + (Y - ZS^{-1}D^T)(A_n - DS^{-1}D^T)^{-1}(Y - ZS^{-1}D^T)^T, \tag{7.8}$$

with $n(A_n - DS^{-1}D^T)^{-1} \to (Y^T F_\kappa Y)^{-1}$. This proves the statements about the covariance, since $Y^T\Lambda_n^{-1}Y = (A_n - DS^{-1}D^T)^{-1}$. For the mean, $nF_n\omega^\star = \Lambda_n\omega^\star - \Lambda_0\omega^\star$ gives

$$\Lambda_n^{-1}\eta_n = \omega^\star - \Lambda_n^{-1}\Lambda_0\omega^\star + \Lambda_n^{-1}\xi_n. \tag{7.9}$$

The second term converges to $ZS^{-1}Z^T\Lambda_0\omega^\star = \Pi_N\omega^\star$. Because $Z^T\xi_n = 0$, the third term equals $(Y - ZS^{-1}D^T)(A_n - DS^{-1}D^T)^{-1}Y^T\xi_n$, which tends to zero since $\xi_n/n \to 0$ almost surely by the strong law. Finally, the Gaussian law with precision Λ_0 conditioned on the affine set $\omega^\star + N$, in the coordinate z of $\omega^\star + Zz$, has precision S and the mean that minimizes $\frac{1}{2}(\omega^\star + Zz)^T\Lambda_0(\omega^\star + Zz)$, which is $z = -S^{-1}Z^T\Lambda_0\omega^\star$. Its mean is ω_∞ and its covariance is $ZS^{-1}Z^T$.

(iii) Here $S = \sigma^{-2}I + \lambda L_N$, and the eigenvalues of S^{-1} are $\sigma^2/(1+\lambda\sigma^2\mu)$ for the eigenvalues $\mu \geq 0$ of L_N. Since W is positive definite, $L_N z = 0$ exactly when $BZz = 0$, so $\ker L_N = Z^T(N \cap \ker B)$ and the rank of L_N is that of BZ.

(iv) From (ii) and $Z^T\Lambda_0\omega^\star = \sigma^{-2}z^\star + \lambda Z^T B^T W B\omega^\star$,

$$Z^T(\omega_\infty - \omega^\star) = -S^{-1}\big(\sigma^{-2}z^\star + \lambda Z^T B^T W B\omega^\star\big), \tag{7.10}$$

which is eq. (7.6) when $B\omega^\star = 0$.

(v) The vector $Z^T B^T W B\omega^\star$ lies in the range of $(W^{1/2}BZ)^T$, which is the range of $L_N = (W^{1/2}BZ)^T(W^{1/2}BZ)$. As $\lambda \to \infty$, $\lambda S^{-1} \to L_N^+$ and $S^{-1} \to 0$ on the range of L_N, while $\sigma^{-2}S^{-1} = I$ on $\ker L_N$. Applying this to eq. (7.10) gives eq. (7.7). □

Without the loop term the recognition law on the invisible directions stays at the prior: messages never move it there, as theorem 3.1 requires. The loop term changes the geometry in which the identified part of the truth is completed, from the anchor to the anchor plus the loop energy, and eq. (7.5) is the resulting projection. In the numerical check, a directed triangle with codes of rank two in $K = 3$ and 10^3 or 10^5 noisy messages per edge, streamed through eq. (7.3), gives mean errors 0.047 and 0.0044 relative to ω_∞, covariance errors 2.0×10^{-3} and 2.0×10^{-5} relative to $ZS^{-1}Z^T$, and relative errors 3.1 and 0.38 percent in the identified block. With $\lambda = 0$ the limit equals the prior on N to 3×10^{-16}, and eqs. (7.6) and (7.7) hold to 2×10^{-14} and to order $1/\lambda$ with a dressed W.

7.3 Filling and detection

Definition 7.3 (Detectable dissonance). The detectable dissonance of a configuration ω is

$$\mathcal{D}(\omega) = \min_{\nu \in N} \|B(\omega + \nu)\|_W^2, \tag{7.11}$$

the least linearized loop energy among the configurations that no message distinguishes from ω.

Proposition 7.4 (Filling and detection are complementary). In the tangent learning model let Q be the W-orthogonal projection of $\mathfrak{g}^{\mathcal{L}}$ onto the W-orthogonal complement of $B(N)$.

(i) $\mathcal{D}(\omega) = \|QB\omega\|_W^2$. In particular $\mathcal{D}$ is constant on cosets of N, so messages determine it.

(ii) With $\Pi_0 = \sigma^{-2} I$, as $\lambda \to \infty$ the limit mean ω_∞ satisfies $B\omega_\infty \to QB\omega^\star$, so its loop energy tends to $\mathcal{D}(\omega^\star)$.

(iii) The loop space $B(X)$ is the W-orthogonal sum of $B(N)$, of dimension $\dim N - \dim(N \cap \ker B)$, and of $B(X) \cap B(N)^{\perp_W}$, of dimension $\operatorname{rank} B - \dim B(N)$. The loop energy splits as $\|B\omega\|_W^2 = \|(I - Q)B\omega\|_W^2 + \|QB\omega\|_W^2$. The first term is the only one that depends on the invisible coordinates, and a dissonance in $B(N)$ is not detectable; the second is a function of the identified coordinates alone.

Proof. (i) Since $B(\omega + Zz) = B\omega + BZz$, the minimum over z leaves the W-orthogonal residual of $B\omega$ from the range of BZ, which is $B(N)$; this residual is $QB\omega$. For $\nu \in N$, $QB\nu = 0$. (ii) By eq. (7.7), $\omega_\infty - \omega^\star$ tends to $-Z(P_{\ker} z^\star + L_N^+ Z^T B^T W B\omega^\star)$. Since $BZP_{\ker} = 0$, $B\omega_\infty$ tends to $(I - BZL_N^+ (BZ)^T W)B\omega^\star$, and $BZ((BZ)^T W BZ)^+ (BZ)^T W$ is the W-orthogonal projection onto the range of BZ, which is $I - Q$. (iii) The dimension of $B(N)$ is the rank of BZ, computed in theorem 7.2(iii). The energy splits because Q is a W-orthogonal projection, and $QB\omega$ does not depend on the component of ω in N by (i). □

The proposition states the cost of the loop prior. Every loop direction in which agreement around loops completes a dictionary is a direction in which the community assumes an agreement it cannot check. If the truth is dissonant there, the recognition law with a strong loop term reports consonance and moves the dissonance into the unread parts of the dictionaries, as the bias term of eq. (7.7). Dissonance is detectable only in the loop directions that the invisible directions cannot reach. In the complete exchange of three agents with codes of rank one in $K = 3$, the invisible directions reach the whole loop space: with a dressed W, the numerical check finds $\mathcal{D}(\omega^\star) = 0$ to 10^{-13} for a

random dissonant truth whose linearized loop energy is 1.02, and the limit mean at $\lambda = 10^8$ has loop energy 4×10^{-12}.

7.4 Which directions loops cannot reach

Proposition 7.5 (The unreachable directions). Let Γ be connected, and suppose that the chains of the walks in $\mathcal{L}$ span the cycle space of Γ, regarded as an undirected multigraph in which the edge $e : j \to i$ is oriented from j to i. Then

$$\ker B = \big\{ (a_i - a_j)_{e : j \to i} : a \in \mathfrak{g}^{\mathcal{V}} \big\}, \tag{7.12}$$

and $N \cap \ker B$ consists of the configurations in eq. (7.12) with $\bar{T}_e(a_i - a_j) = 0$ on every edge $e : j \to i$. For $\mathfrak{g} = \mathfrak{gl}(K, \mathbb{R})$,

$$\begin{aligned} \dim(N \cap \ker B) &= K\big(\dim H^0 - K\big), \\ H^0 &= \big\{ u \in (\mathbb{R}^K)^{\mathcal{V}} : \bar{T}_e(u_i - u_j) = 0 \text{ for every edge } e : j \to i \big\}. \end{aligned} \tag{7.13}$$

Proof. The value $(B\omega)_\gamma$ is the pairing of ω, read as a $\mathfrak{g}$-valued cochain on edges, with the chain of γ, and that chain is a cycle because a closed walk leaves each vertex as often as it enters it. If these chains span the cycle space, $\omega \in \ker B$ exactly when ω vanishes on every cycle. Choose a spanning tree and a root, let a_v be the sum of ω along the tree path from the root to v, each edge taken with the sign of its orientation, and put a equal to zero at the root. Then $\omega_e - (a_i - a_j)$ vanishes on tree edges and, on any other edge, equals the value of ω on its fundamental cycle, which is zero. Conversely every configuration of the form eq. (7.12) vanishes on cycles. The condition for membership in N is the definition of N_e. For $\mathfrak{gl}(K, \mathbb{R})$ the condition $\bar{T}_e(a_i - a_j) = 0$ holds column by column, so $a \mapsto (a_i - a_j)_e$ maps the families whose K columns lie in H^0, a space of dimension $K \dim H^0$, onto $N \cap \ker B$. Its kernel consists of the constant families, of dimension K^2, because Γ is connected. □

The configurations in eq. (7.12) are the infinitesimal pure-gauge configurations, what the dictionaries would become if each agent's reference coordinates were moved by e^{a_v}. The space $N \cap \ker B$ consists of the re-readings of this kind that no attending code registers. They create no holonomy, so no loop term reaches them, and no message reveals them. The space H^0 is the space of global sections of a cellular sheaf on Γ whose stalk at an agent is $\mathbb{R}^K$, whose stalk at an edge is the message space, and whose two restriction maps at an edge are both the attending code; it is a sheaf of the kind that Hansen and Ghrist [9] use for discourse, arising here from identifiability.

Corollary 7.6 (Complete exchange). Let every pair of agents exchange in both directions, and let $\mathcal{L}$ contain every two-cycle and every directed triangle. Then the hypothesis of proposition 7.5 holds and

$$H^0 = \big\{ u : u_i - u_j \in \ker \bar{T}_{j \to i} \cap \ker \bar{T}_{i \to j} \text{ for every pair } \{i, j\} \big\}, \tag{7.14}$$

where $\bar{T}_{j \to i}$ is the code on the edge $j \to i$. If the pairs with $\ker \bar{T}_{j \to i} \cap \ker \bar{T}_{i \to j} = 0$ connect all agents, then H^0 consists of constant sections, $N \cap \ker B = 0$, and for a flat truth the recognition law recovers every dictionary in the limits $n \to \infty$ and then $\lambda \to \infty$.

Proof. Fix an agent 0. The edges $0 \to i$ form a spanning tree. The fundamental cycle of $i \to 0$ is the two-cycle through 0 and i, and that of $j \to i$ with $i, j \neq 0$ is the directed triangle $0 \to j \to i \to 0$ minus the two-cycle through 0 and i. The two edges between i and j give $\bar{T}_{j\to i}(u_i - u_j) = 0$ and $\bar{T}_{i\to j}(u_j - u_i) = 0$. If the pairs with trivial intersection connect the agents, u is constant along each of them, hence constant. The last statement is theorem 7.2(iv). □

The sufficient condition is not necessary: loops through a third agent can fix what no pair can. In $K = 3$ with codes of rank one, the kernels of every pair meet in a line, and the numerical check finds that the three two-cycles alone leave $9 = \sum_{\text{pairs}} K \dim(\ker \cap \ker)$ unreachable dimensions out of 36. Adding the two directed triangles leaves none, because $u_1 - u_0$, $u_2 - u_1$ and $u_0 - u_2$ would have to lie on three lines in general position and sum to zero. If one direction k is read by no code, the count is $K(\dim H^0 - K) = 6$, from the shifts of each agent along k. A directed triangle alone, in which each agent hears one other through a code of rank one, leaves 9 of 18. Every count agrees with eq. (7.13).

Proposition 7.7 (Reciprocal pair). Let two agents exchange in both directions with $\mathcal{L}$ the two-cycle, W the Frobenius norm and $\mathfrak{g} = \mathfrak{gl}(K, \mathbb{R})$. Write $\bar{T}_1$ for the code on $1 \to 0$ and $\bar{T}_0$ for the code on $0 \to 1$, $\mathcal{K}_a = \ker \bar{T}_a$, $\mathcal{R}_a = \mathcal{K}_a^\perp$ for the row space of $\bar{T}_a$, and $P_\cap$ for the orthogonal projection onto $\mathcal{R}_0 \cap \mathcal{R}_1$. Then

$$\begin{aligned} \dim B(N) &= K \dim(\mathcal{K}_0 + \mathcal{K}_1), \qquad \dim(N \cap \ker B) = K \dim(\mathcal{K}_0 \cap \mathcal{K}_1), \\ \operatorname{rank} B - \dim B(N) &= K \dim(\mathcal{R}_0 \cap \mathcal{R}_1), \end{aligned} \tag{7.15}$$

and $\mathcal{D}(\omega) = \|P_\cap(\omega_{01} + \omega_{10})\|_F^2$.

Proof. N consists of the pairs (ν_{01}, ν_{10}) with the range of ν_{01} in $\mathcal{K}_1$ and the range of ν_{10} in $\mathcal{K}_0$, and $B\nu = \nu_{01} + \nu_{10}$. Hence $B(N)$ is the set of matrices with range in $\mathcal{K}_0 + \mathcal{K}_1$, of dimension $K \dim(\mathcal{K}_0 + \mathcal{K}_1)$. Its Frobenius-orthogonal complement in $B(X) = \mathfrak{gl}(K, \mathbb{R})$ is the set of matrices with range in $(\mathcal{K}_0 + \mathcal{K}_1)^\perp = \mathcal{R}_0 \cap \mathcal{R}_1$, and the orthogonal projection onto it is $X \mapsto P_\cap X$. Thus $Q = P_\cap$ and proposition 7.4(i) gives $\mathcal{D}$. The space $N \cap \ker B$ consists of the pairs with $\nu_{01} = -\nu_{10}$ and range in $\mathcal{K}_0 \cap \mathcal{K}_1$. □

A pair detects its mutual misunderstanding only in what both can say, repairs its dictionaries through reciprocity in what at least one of them can say, and leaves what neither can say at the prior. When the two row spaces are independent and together have dimension K, everything invisible is repaired and no misunderstanding is detectable: complete repair costs complete trust. The numerical check at $K = 3$ gives the predicted triples $(6, 0, 3)$ of filled, unreachable and detectable dimensions for two codes of rank two and $(9, 3, 0)$ for two codes of rank one, and $\mathcal{D}$ agrees with the projection formula to 10^{-14}.

7.5 Without linearization

The tangent model is exact about which configurations remain consistent with the messages and with flat loops. The following two statements hold for the grounded model eq. (3.1) itself, in the constraint form of the loop term.

Proposition 7.8 (Exact reciprocal identification). Let $G = \mathrm{GL}(K, \mathbb{R})$ and let the true dictionaries of two agents be reciprocal, $\bar\theta^\star_{10} = (\bar\theta^\star_{01})^{-1}$. The pairs $(\bar\theta_{01}, \bar\theta_{10})$ that reproduce the message laws of both edges for every referent and satisfy $\bar\theta_{01}\bar\theta_{10} = I$ are exactly

$$\bar\theta_{01} = \bar\theta^\star_{01} s, \qquad \bar\theta_{10} = s^{-1}\bar\theta^\star_{10}, \qquad s \in \mathrm{Stab}(\bar T_1) \cap \mathrm{Stab}(\bar T_0 \bar\theta^\star_{01}). \tag{7.16}$$

The pair is determined uniquely exactly when the stacked map $(\bar T_1; \bar T_0\bar\theta^\star_{01})$ is injective; otherwise the solutions form a set of dimension $K \dim\ker(\bar T_1; \bar T_0\bar\theta^\star_{01})$.

Proof. By theorem 3.1(i), the edge $1 \to 0$ reproduces its law exactly when $\bar\theta_{01} = \bar\theta^\star_{01} s$ with $s \in \mathrm{Stab}(\bar T_1)$. Reciprocity forces $\bar\theta_{10} = s^{-1}\bar\theta^\star_{10}$, and the edge $0 \to 1$ reproduces its law exactly when $(\bar\theta^\star_{10})^{-1} s^{-1} \bar\theta^\star_{10} = \bar\theta^\star_{01} s^{-1} (\bar\theta^\star_{01})^{-1} \in \mathrm{Stab}(\bar T_0)$, that is, when $\bar T_0 \bar\theta^\star_{01} s^{-1} = \bar T_0 \bar\theta^\star_{01}$. The intersection of the two stabilizers is the stabilizer of the stacked map, the set of s with $s - I$ mapping $\mathbb{R}^K$ into its kernel, which is open in an affine space of dimension K times the dimension of that kernel. □

The map $\bar T_0\bar\theta^\star_{01}$ is agent 0's code expressed in agent 1's reference coordinates, so the pair is identified exactly when, between them, the two agents can say everything. In the numerical check, with codes of rank one in $K = 3$ and a true dictionary far from the identity, a finite element of the residual set reproduces both message laws to 7×10^{-16}, is reciprocal to 4×10^{-16}, and differs from the truth by 1.86 in the largest entry.

Proposition 7.9 (Exact flat identification). Let $G = \mathrm{GL}(K, \mathbb{R})$ and Γ be connected, and suppose that every configuration whose holonomies on $\mathcal{L}$ are the identity is pure gauge, $\bar\theta_{ij} = k_i k_j^{-1}$ for some $k \in G^{\mathcal{V}}$. Let the truth be pure gauge, $\bar\theta^\star_{ij} = g_i g_j^{-1}$, and put $\bar T'_e = \bar T_e g_j$ for $e : j \to i$. The configurations that reproduce every message law and have identity holonomy on $\mathcal{L}$ are exactly

$$\bar\theta_{ij} = g_i u_i u_j^{-1} g_j^{-1}, \qquad u \in G^{\mathcal{V}}, \qquad \bar T'_e(u_i - u_j) = 0 \text{ for every edge } e : j \to i, \tag{7.17}$$

and two families u, u' give the same configuration exactly when $u'_v = u_v c$ for all v and one $c \in G$. The truth is the only such configuration exactly when the space H^0 of eq. (7.13), formed with the codes $\bar T'_e$, consists of constant sections. The hypothesis holds for the complete exchange with every two-cycle and every directed triangle.

Proof. By hypothesis $\bar\theta_{ij} = k_i k_j^{-1}$; write $k_v = g_v u_v$. By theorem 3.1(i), the edge e reproduces its law exactly when $(\bar\theta^\star_{ij})^{-1}\bar\theta_{ij} = g_j u_i u_j^{-1} g_j^{-1} \in \mathrm{Stab}(\bar T_e)$, that is, when $\bar T'_e u_i u_j^{-1} = \bar T'_e$, that is, when $\bar T'_e u_i = \bar T'_e u_j$. Two families give the same configuration exactly when $(u'_i)^{-1}u_i = (u'_j)^{-1}u_j$ on every edge, which on a connected graph means $u' = uc$. If H^0 consists of constant sections, every column of u is constant, so $u_v = u_0$ for all v and the configuration is the truth. If $s \in H^0$ is not constant, $u_v = I + t s_v w^T$ with $w \neq 0$ and small $t \neq 0$ is an invertible family satisfying the condition, and it is not of the form $(c)_v$, so its configuration differs from the truth. For the complete exchange put $k_0 = I$ and $k_i = \bar\theta_{i0}$. The two-cycle through 0 and i gives $\bar\theta_{0i} = k_i^{-1}$, and the directed triangle $0 \to j \to i \to 0$, whose holonomy is $\bar\theta_{0i}\bar\theta_{ij}\bar\theta_{j0}$, gives $\bar\theta_{ij} = \bar\theta_{0i}^{-1}\bar\theta_{j0}^{-1} = k_i k_j^{-1}$. □

The residual set of eq. (7.17) has dimension $K(\dim H^0 - K)$ at the truth, the count of proposition 7.5. In the numerical check, the complete exchange of three agents with a direction read by no code in the gauge of the truth, and a truth far from the identity, has a finite element of the residual set that

reproduces all six message laws to 7×10^{-15}, has holonomy equal to the identity to 4×10^{-15} on all five loops, and differs from the truth by 3.7 in the largest entry.

7.6 Receiver-owned recognition laws

Remark 7.10 (Ownership and the loop term). In the per-agent picture of the main manuscript each receiver infers only its own dictionaries, so its recognition law is one factor $q_i(\omega_{i\cdot})$ of a law factorized over receivers. A two-cycle couples two receivers and a directed triangle three, so the loop term couples the factors. In the tangent model the factorized law closest to the posterior in $D_{\mathrm{KL}}(q\|\text{posterior})$ has the exact posterior means and, as covariances, the inverses of the diagonal blocks of the posterior precision [5, Section 10.1.2]. These are smaller in the Loewner order than the exact marginal covariances. In the Gaussian tangent model, ownership therefore leaves the estimates of the dictionaries unchanged and makes each receiver overconfident about them. Exact means are a feature of Gaussian targets: for a non-Gaussian posterior a factorized law generally has means different from the exact ones, so for general laws ownership can change the estimates as well. To update its factor, receiver i needs the current means of the dictionaries that other agents hold along its loops: a loop prior can be used only if agents communicate about how they read each other, in addition to communicating about the world. In the numerical check, coordinate ascent over the three receivers of the complete exchange reproduces the exact means to 10^{-14}, the differences between exact and factorized block covariances are positive semidefinite, and the total variance is 4.59 against the exact 7.75.

Chapter 8

Next steps

Open problem 8.1 (Learning the dictionaries beyond the tangent model). Theorem 7.2 and proposition 7.4 settle the learning problem in the tangent model, and propositions 7.8 and 7.9 settle the constraint form of the loop term for a flat truth without linearization. Three parts remain. First, for the full grounded model eq. (3.1) with a prior on ω, show that the recognition law converges to the prior conditioned on the fiber $\prod_e \bar{\theta}_e^\star \operatorname{Stab}(\bar{T}_e)$, which is not linear in the coordinates ω, and determine the selection within the fiber for a dissonant truth and a finite loop weight. Second, replace the positive-definite loop forms by the law-based loop defects of the main manuscript. These are only semidefinite and do not register holonomy that fixes the base law, so more directions should remain unreachable. Third, treat latent routing labels, which lower the information below the routed sum of corollary 4.2, and loop weights built from the routing weights. Fourth, replace the Gaussian anchor and noise by general laws: for a location family of noise and a non-Gaussian prior on ω, determine the limit of the recognition law on the fiber, which is no longer a conditioned Gaussian, and whether the filled directions are still recovered exactly as the loop weight grows.

Open problem 8.2 (Codes as part of the species model). Let codes be learned as well, so that a shared language is a joint fixed point of code and dictionary learning. The expression-map dynamics of Hansen and Ghrist [9] is the vector-valued analogue on a sheaf. Determine when the learned codes of a species become related by its frame map, and when learning drives code-visible dissonance to zero while leaving dissonance in unread directions.

Open problem 8.3 (Dictionaries in the free-energy pictures). Place receiver-owned dictionaries and codes in the per-agent generative model of the main manuscript, where each receiver infers its own dictionaries, and in a global joint, where all dictionaries are variables of one model. Compare the resulting dynamics with the reciprocal flow, and decide whether consensus dynamics reduces visible dissonance and pushes holonomies toward the stabilizers of the codes.

Open problem 8.4 (Beyond Gaussian codes). Propositions 2.2 and 6.1 and corollary 2.4 already hold for general laws and measurable codes, including codes into finite alphabets such as words, and the consensus theorem holds for laws as well as means. Remark 3.2 extends identifiability to every location family of noise. Extend the identifiability, learning and dissonance results to codes whose message laws are not a location family, such as codes into finite alphabets, where the Fisher form is replaced by the information in the induced discrete channel and the stabilizer by the subgroup that leaves the channel unchanged, and to agents of different kinds, whose dictionaries are not invertible.

Related work

The ingredients appear separately in the literature; their combination was not found in a standard search, which is not exhaustive. Opinion dynamics on discourse sheaves [9] gives agents private opinion spaces and linear expression maps into shared discourse spaces, with dynamics that can also evolve the expression maps; opinions are vectors, and there is no receiver reconstruction, gauge group or holonomy. A cognitive-geometric model of interpretation [1] uses linear interpretation maps between agents' value spaces, with beliefs lost in their null spaces and composite maps along chains. Communication without a shared reference frame treats unknown frame changes as gauge data [3, 18]. The meeting of minds in conceptual spaces models meaning as maps between individual spaces with agreement as a fixed point [20]. The relativistic theory of consciousness [13] holds that an observer reading another system's representation measures only its physical coordinates and constructs a new representation of its own; the code-then-reconstruction chain eq. (1) is a formal version of that measurement, and the identity dictionary between equivalent systems is the transformation it uses for that case. Communication as active inference with a shared generative model is developed by Friston and Frith [7] and Friston et al. [8], and language as efficient compression by Zaslavsky et al. [22], in the tradition of Shannon [15], the information bottleneck [17] and coding with side information [21].

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